

Amaliy mashgʻulot

Sarvar Amirov

March 2026

1 To'g'ri chiziqli tekis harakat

1. Qanday harakat to'g'ri chiziqli tekis harakat bo'ladi?
 - A) Teng vaqt oraliqlarida bir xil masofaga ko'chuvchi va trayektoriyasi to'g'ri chiziqdan iborat bo'lgan jism harakati
 - B) Trayektoriyasi to'g'ri chiziqdan iborat harakat
 - C) Hech qanday tezlanishga ega bo'lmagan harakat
 - D) Teng vaqt oraliqlarida teng masofalar o'tuvchi jism harakati
2. To'g'ri chiziqli tekis harakatda tezlik vektorining ...
 - A) moduli tekis oshib, yo'nalishi o'zgarmaydi
 - B) moduli va yo'nalishi o'zgaras saqlanadi
 - C) moduli va yo'nalishi o'zgarib turadi
 - D) moduli o'zgaras bo'lib, yo'nalishi uzluksiz o'zgarib turadi
3. 918 km/h tezlik necha m/s bo'ladi?
 - A) 260
 - B) 240
 - C) 250
 - D) 255
4. Uzunligi 9 m bo'lgan ingichka ip boshidan jo'nagan chumoli ip oxiriga kelguncha 5 minut o'tdi. Chumolining tezligini toping (cm/s).
 - A) 1,8
 - B) 3
 - C) 3,5
 - D) 4,5
5. Uzunligi 200 m bo'lgan poyezd 10 m/s tezlik bilan tekis harakatlanib, uzunligi 300 m bo'lgan tunnelga kirib bormoqda. Necha sekunddan keyin poyezd tunneldan butunlay chiqib ketadi?
 - A) 50
 - B) 30
 - C) 25
 - D) 20
 - E) 10
6. Birinchi avtomobil $4S$ masofani $2t$ vaqtda $2v$ tezlik bilan bosib o'tdi. Ikkinchi avtomobil $3S$ yo'lni $2t$ vaqtda bosib o'tsa, uning tezligini toping.
 - A) $2.25v$
 - B) $2.5v$
 - C) $1.5v$
 - D) $0.5v$
 - E) v
7. Magnitofon o'ragichi 4 m/s tezlik bilan 50 s da tasmani o'rab bo'ldi. Agar o'ragichning boshlang'ich radiusi 3 cm, oxirgi radiusi 7 cm bo'lsa, tasmaning qalinligini toping (mm). $\pi = 3$ deb oling.
 - A) 0.5
 - B) 0.06

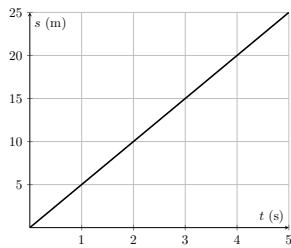
- C) 0.1
D) 0.09
E) 0.08
8. Uzunligi 300 m bo'lgan, 36 km/h tezlik bilan harakatlanayotgan mashinalar qatori 60 m uzunlikdagi ko'prikdan necha sekunda o'tib ketadi?
- A) 30
B) 60
C) 90
D) 120
9. Piyoda A punktdan 30 km uzoqlashgach, piyodanikidan 5 marta katta tezlik bilan velosipedchi yo'lga chiqdi. Agar ular B punktga bir paytda kirib borgan bo'lsalar, punktlar orasidagi masofa qanday (km)?
- A) 20
B) 22.4
C) 25
D) 37.5
E) 36
10. A va B punktlardan bir-biriga qarab ikki velosipedchi yo'lga chiqdi. Ular B punktga 30 km qolganda uchrashishadi. Manzilga borib qaytishda esa A punktdan 18 km uzoqlikda uchrashishdi. Punktlar orasidagi masofani toping (km).
- A) 48
B) 60
C) 68
D) 72
E) 78
11. Ikki jismning harakat tenglamalari $x_1 = 25 + t$, $x_2 = 85 - 0.5t$ ko'rinishga ega. Ularning uchrashish vaqti va joyi aniqlansin.
- A) 40 s dan so'ng, koordinata boshidan 65 m masofada
B) 18 s dan so'ng, koordinata boshidan 114 m masofada
C) 10 s dan so'ng, koordinata boshidan 5 m masofada
D) 30 s dan so'ng, koordinata boshidan 60 m masofada
E) TJY
12. Harakat tenglamalari $x_1 = 2 + 3t$, $x_2 = 1 - 2t$ bo'lgan moddiy nuqtalar vaqt boshidan necha sekund o'tganda uchrashadilar?
- A) ular uchrashmaydilar
B) 1
C) 2
D) 3
E) 4
13. Yerdan 135 m balandlikdan uchib ketayotgan dronni razvedkachi durbin yordamida kuzatmoqda. Bir paytda dron vertikalga nisbatan 45° burchak ostida ko'rindi. Shundan 6 s o'tgach 53° burchak ostida ko'rindi. Dronning tezligini toping (km/h). $\sin 53^\circ = 0.8$, $\cos 53^\circ = 0.6$.
- A) 15

- B) 34
C) 48
D) 27
E) 9
14. Jism gorizont bilan 60° burchak hosil qilgan qiyalikdan 20 m/s tezlik bilan yuqoriga tortilmoqda. Jismning vertikal va gorizontaal yo'nalishlardagi tezliklari qanday (m/s)?
A) 17.3; 10
B) 10; 14.1
C) 10; 17.3
D) 14.1; 10
E) TJY
15. Moddiy nuqta XOY tekisligida $x = 11 + 1.5t$, $y = 3 + 2t$ tenglamalar bilan harakat qilmoqda. Tezligini aniqlang (m/s).
A) 4
B) 3.5
C) 5
D) 5.25
E) 2.5
16. Moddiy nuqtaning harakati $x = 7 + 12t$, $y = 5 + 16t$ tenglamalar bilan berilgan. Dastlabki 3 s ichida qancha ko'chadi (m)?
A) 14
B) 19
C) 10
D) 13
E) 60
17. Moddiy nuqta $x = 2 + t$, $y = 3 + 2t$ tenglamalar bilan harakat qiladi. Trayektoriya tenglamasini toping.
A) $y = 2x$
B) $y = 2x - 1$
C) $y = 2x + 2$
D) $y = x + 1$
E) $y = 2x + 3$

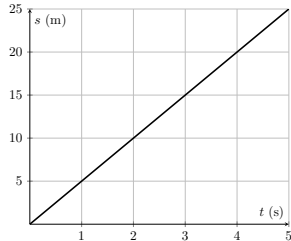
1a	2b	3a	4b	5a	6c	7e	8c	9d	10d
11a	12a	13d	14a	15e	16e	17b			

2 To'g'ri chiziqli tekis harakatni grafik ravishda tasvirlash

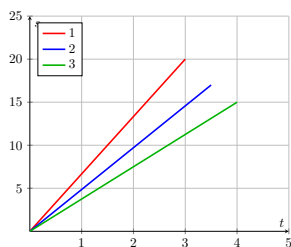
1. Rasmda bola bosib o'tgan yo'ning vaqtga bog'lanish grafigi keltirilgan. Bolaning $t_1 = 1$ s va $t_2 = 5$ s vaqt oralig'ida bosib o'tgan yo'lini aniqlang.



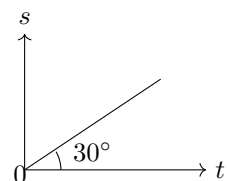
- A) 10 m
B) 20 m
C) 5 m
D) 15 m
2. Rasmda velosipedchi o'tgan yo'ning vaqtga bog'lanish grafigi keltirilgan. Uning $t = 4.28$ s vaqt momentidagi tezligi qanday (m/s)?



- A) 5
B) 12
C) 10
D) 6
3. Rasmda uchta moddiy nuqta bosib o'tgan yo'ning vaqtga bog'lanish grafigi berilgan. Qaysi moddiy nuqtaning tezligi eng katta?

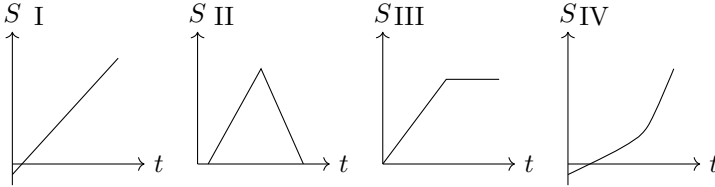


- A) 1
B) 2
C) 3
D) hammasiniki bir xil
4. Quyidagi grafikdan foydalanib jismning harakat tezligini aniqlang (m/s).



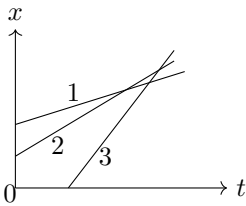
- A) $\sqrt{3}$
- B) $\frac{\sqrt{3}}{3}$
- C) 3
- D) $\frac{1}{2}$

5. Qaysi grafik jism o'tgan yo'lining vaqtga bog'lanishini ifodalay olmaydi?



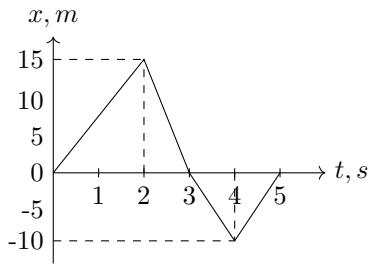
- A) II, III
- B) I, II, III
- C) III
- D) I, IV

6. Chizmada uchta moddiy nuqtaning harakat grafiklari berilgan. Ularning tezliklari haqida nima deyish mumkin?



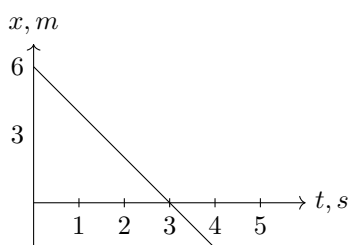
- A) $v_1 = v_2 < v_3$
- B) $v_1 > v_2 > v_3$
- C) $v_1 = v_2 = v_3$
- D) $v_1 = v_2 > v_3$

7. Grafikdan foydalanib OX o'q bo'ylab harakat qilayotgan moddiy nuqtaning 5 s da bosib o'tgan yo'lini aniqlang (m).



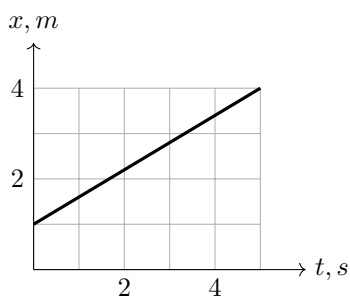
- A) 50
- B) 32.5
- C) 0
- D) 10

8. Rasmda jism koordinatasining vaqtga bog'liqlik grafigi berilgan. Jismning harakat tenglamasini yozing.



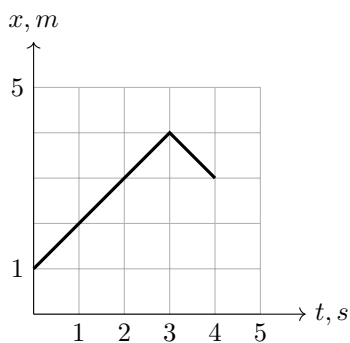
- A) $x = -2 + 4t$
- B) $x = 6 + 3t$
- C) $x = -6 + 3t$
- D) $x = 6 - 2t$

9. Rasmda jismning harakat grafigi keltirilgan. $t = 3$ s paytdagi jismning koordinatasini aniqlang (m).



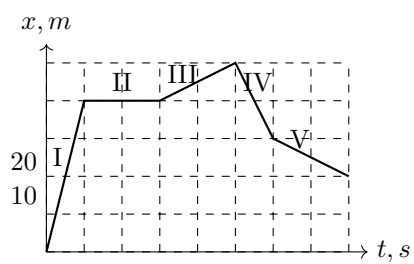
- A) 1.8
- B) 1.6
- C) 2.8
- D) 1.2

10. Rasmda jism koordinatasining vaqtga bog'lanish grafigi tasvirlangan. Jismning $(1 - 4)$ s vaqt oralig'idagi ko'chishini va yo'lini aniqlang (m).



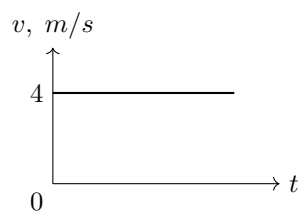
- A) 1; 4
- B) 2; 4
- C) 1; 3
- D) 2; 3

11. Rasmda jismning harakat grafigi ko'rsatilgan. Grafikning qaysi qismida jismning tezligi eng katta bo'lgan?



- A) III
- B) II
- C) I
- D) IV

12. Chizmada keltirilgan tezlik grafigidan foydalanib X o'qi bo'ylab harakatlanayotgan jismning harakat tenglamasini tuzing. Boshlang'ich koordinata -10 m.



- A) $x = -10 + 4t$
- B) $x = -10 + 2t^2$
- C) $x = -10t + 4t^2$
- D) $x = -10 + 2t$

1b	2a	3a	4b	5c	6c	7a	8d	9d	10b
11c	12a								